

Final Project Report

Project Title: WAIN, an AI-Powered Riyadh Place Recommendation Assistant

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Cohort / Track: Agentic AI Bootcamp - Cohort 2

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Project Links: [[GitHub Repository](#)] | [[Live Demo / Deployment Link](#)]

1. Executive Summary & Problem Statement

- **Problem Statement:** In Riyadh, users looking for restaurants, cafés, activities, and places to visit often need to browse through many listings and manually apply different filters to find options that match their preferences, such as budget, location, category, rating, and activity type. This process can be time-consuming and inconvenient.
- **Target Audience:** WAIN is designed for people looking for restaurants, cafés, activities, attractions, and other places in Riyadh. It helps users find relevant options based on their specific preferences without requiring them to manually search through multiple listings or apply several filters.
- **Proposed Solution:** WAIN is an AI-powered place recommendation agent designed specifically for Riyadh. It understands users' requests in natural language, extracts their preferences and constraints, retrieves relevant places from a PostgreSQL database, ranks the retrieved options based on the user's preferences, and generates a personalized response with suitable recommendations.

2. Core Features & System Architecture

➤ Key Features:

- **Natural-Language Search:** Users can describe what they are looking for in natural language, including preferences such as budget, category, subcategory, rating, location, activity type.

- **AI-Based Preference Extraction:** WAIN uses an LLM to interpret the user's request and extract relevant preferences and constraints from the natural-language input.
- **Personalized Retrieval & Ranking:** WAIN retrieves matching places from the dataset and ranks the results according to the user's specified preferences.
- **Clear Recommendations:** WAIN presents the recommended places in a clear and user-friendly format, including key information such as place name, category, location, rating, and estimated cost, with map links for navigation.

➤ **Tech Stack:**

- **Frontend / User Interface:** Streamlit
- **Backend / Business Logic:** Python
- **AI / LLM Layer:** OpenAI API
- **Database / Storage:** PostgreSQL
- **Development & Version Control:** PyCharm, Git, GitHub

3. Future Scope & Conclusion

➤ **Short-Term Enhancements:**

- Add Arabic language support to allow users to interact with WAIN in both Arabic and English.
- Expand and improve the dataset by adding more restaurants, cafés, attractions, activities, and places across Riyadh.

➤ **Long-Term Potential:**

- Expand WAIN beyond Riyadh to cover other cities across Saudi Arabia.
- Continuously grow the database to provide a wider range of places, activities, and experiences.
- Develop WAIN into a full-scale commercial platform for personalized local place recommendations.